Modelling of a Ball Mill as a Custom Unit Operation in DWSIM

Process Simulation using Python-Based Custom Unit Operation

Abstract

This work presents the development and implementation of a ball mill as a custom Python-scripted unit operation (UO) within the open-source process simulator DWSIM. Ball milling is a widely employed size-reduction operation in the mineral processing, cement, pharmaceutical, and chemical industries. Despite its industrial importance, DWSIM does not natively include a ball mill unit operation with rigorous particle size distribution (PSD) tracking. The present work addresses this gap by integrating established comminution theory directly into the DWSIM simulation environment through its built-in Python scripting interface.

The mathematical model is built on two complementary frameworks. First, Austin's Population Balance Model (PBM) with first-order breakage kinetics is employed to simulate the evolution of the particle size distribution through the mill. The model incorporates the selection function S_i, which quantifies the rate of breakage for each particle size class, and the breakage distribution function B_{ij}, which describes the redistribution of broken particles into finer size classes. A mill diameter correction factor is applied to the selection function following Austin et al. (1984). Second, Bond's Third Law of Comminution is used to estimate the specific energy consumption of the grinding operation based on the 80th-percentile passing sizes of the feed (F_{80}) and product (P_{80}) streams, with the Bond Work Index (W_i) as the primary material-specific parameter.

The custom UO accepts a single inlet material stream and produces a single outlet stream with updated temperature, accounting for the thermal energy imparted by grinding. The residence time distribution inside the mill can be modelled either as a Continuous Stirred Tank Reactor (CSTR), solved algebraically at steady state, or as a Plug Flow Reactor (PFR), solved via Euler time integration over the mean residence time. Mill power draw is estimated using the Hogg–Fuerstenau model as a function of mill diameter, length, ball filling fraction, and rotational speed expressed as a fraction of critical speed. User-defined parameters - including particle size class boundaries, feed PSD, residence time, Bond Work Index, and Austin model coefficients - are supplied through DWSIM's Input Variables and Input Strings parameter tables, following the established convention of existing DWSIM custom UO scripts.

The model was successfully integrated into DWSIM and executed without errors after resolving Python–.NET interoperability issues specific to the DWSIM scripting environment, including the absence of the *ScriptVariables* object and the requirement to wrap scalar .NET properties in Python lists prior to stream assignment. Diagnostic outputs - including F_{80}, P_{80}, specific energy consumption, mill power draw, product temperature rise, and the full product PSD - are printed to the DWSIM flowsheet console upon each calculation. Validation of the model is recommended against tabulated batch grinding data from Austin et al. (1984) and standard Bond grindability test benchmarks.

The developed custom UO demonstrates that rigorous comminution models can be embedded within DWSIM's process simulation framework, enabling engineers to incorporate ball milling into full flowsheet simulations involving solid–liquid separation, classification, and recycle loops. The modular structure of the script allows straightforward extension to multi-compartment mills, variable residence time distributions, or slurry rheology corrections in future work.

Keywords

- Ball mill; size reduction; comminution
- Population Balance Model; Austin breakage kinetics
- Bond's Law; Work Index; specific energy
- DWSIM; custom unit operation; Python scripting
- Particle size distribution; process simulation

Key References

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